

A Research Analysis on Security Challenges and Risk Mitigation Strategies in Cloud Computing Environments – AWS

AWS-Based Research Analysis on Security Challenges and Risk Mitigation Strategies in Cloud Computing Environments

Submitted in partial fulfilment of the requirements for the award of the degree of

Master of Computer Applications (MCA)

By

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Institute Name: Chandigarh University

Academic Year: 2024-25

CERTIFICATE

Paste the Qollabb project certificate here.

DECLARATION

I, Shripad Ghole, hereby solemnly declare that the project report titled “Based Research Analysis on Security Challenges and Risk Mitigation Strategies in Cloud Computing Environments” submitted in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications (MCA) is my original work.

I further declare that:

- This project has been carried out by me during the academic year 2024-2025 under the supervision of Kunwar Saurabh Bisen
- The work has not been submitted previously to any other university, institution, or examination body.
- All references and sources have been properly acknowledged.
- The project implementation and analysis are authentic to the best of my knowledge.

Place: Pune, Maharashtra

Date: 15-05-2026

Student Signature: Shripad Ghole

Student Name: Shripad Ghole

Enrollment No: O24MCA160241

ACKNOWLEDGEMENT

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I extend my appreciation to Qollabb for providing the opportunity to work on this industry-oriented project and gain practical knowledge in cloud security implementation and analysis using AWS cloud services.

Finally, I would like to thank all individuals and online documentation resources that helped me complete this project successfully.

ABSTRACT

Cloud computing has become one of the most widely adopted technologies due to its scalability, flexibility, and cost efficiency. However, increasing dependence on cloud infrastructure has also introduced several security challenges such as data breaches, misconfiguration, unauthorized access, and insecure APIs.

This project focuses on analysing cloud security threats and implementing risk mitigation strategies using Amazon Web Services (AWS). The project includes practical implementation of cloud security controls using services such as Amazon S3, AWS IAM, AWS CloudTrail, Amazon GuardDuty, and AWS Config.

The project demonstrates security threat identification, monitoring, logging, encryption, access control, and security analysis through practical AWS implementation. Various security controls including Multi-Factor Authentication (MFA), least privilege access, encryption, and continuous monitoring were implemented to improve cloud security posture.

The project also includes testing, threat analysis, and future cloud security trends such as AI-based threat detection and Zero Trust Architecture.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AWS	Amazon Web Services
IAM	Identity and Access Management
MFA	Multi-Factor Authentication
API	Application Programming Interface
DFD	Data Flow Diagram
AI	Artificial Intelligence
TLS	Transport Layer Security
S3	Simple Storage Service

INTRODUCTION

Background of the Project:-

Cloud computing is one of the most important technologies in modern IT infrastructure. Organizations use cloud services to store data, host applications, and manage business operations due to advantages such as scalability, flexibility, and cost reduction.

However, cloud environments are highly vulnerable to security threats including data breaches, misconfiguration, insider threats, insecure APIs, and unauthorized access. Many organizations fail to implement proper cloud security practices, leading to financial and reputational losses.

This project focuses on analyzing cloud security challenges and implementing practical mitigation strategies using AWS cloud services.

Problem Statement:-

Organizations using cloud platforms often face multiple security risks due to weak access management, public cloud storage exposure, insufficient monitoring, and lack of encryption.

Major problems include:

- Publicly accessible cloud storage
- Weak IAM permissions
- Lack of continuous monitoring
- Unauthorized access risks
- Misconfiguration vulnerabilities

Objectives of the Project:-

The main objectives of this project are:

- To analyze cloud security threats
- To implement AWS cloud security controls
- To monitor cloud activities and logs
- To apply risk mitigation strategies
- To evaluate security frameworks
- To analyze future cloud security trends

Scope of the Project:-

The project scope includes:

- AWS cloud implementation
- Amazon S3 security
- IAM configuration
- MFA implementation
- CloudTrail logging
- GuardDuty monitoring
- AWS Config compliance monitoring
- Security analysis and testing

Existing System Overview:-

Traditional cloud environments often rely on manual security management and lack automated monitoring systems. Many organizations fail to implement proper access controls and encryption mechanisms.

Existing systems face issues such as:

- Poor visibility into cloud activities
- Weak authentication methods
- Insufficient logging and monitoring
- Delayed threat detection

Proposed System Overview:-

The proposed system uses AWS cloud services to implement cloud security monitoring and risk mitigation.

The system includes:

- Secure cloud storage
- IAM access management
- Multi-Factor Authentication
- CloudTrail logging
- GuardDuty threat detection
- AWS Config compliance monitoring

Technologies Used:-

- Amazon Web Services (AWS)
- Amazon S3
- AWS IAM
- AWS CloudTrail
- Amazon GuardDuty
- AWS Config
- MFA Authentication

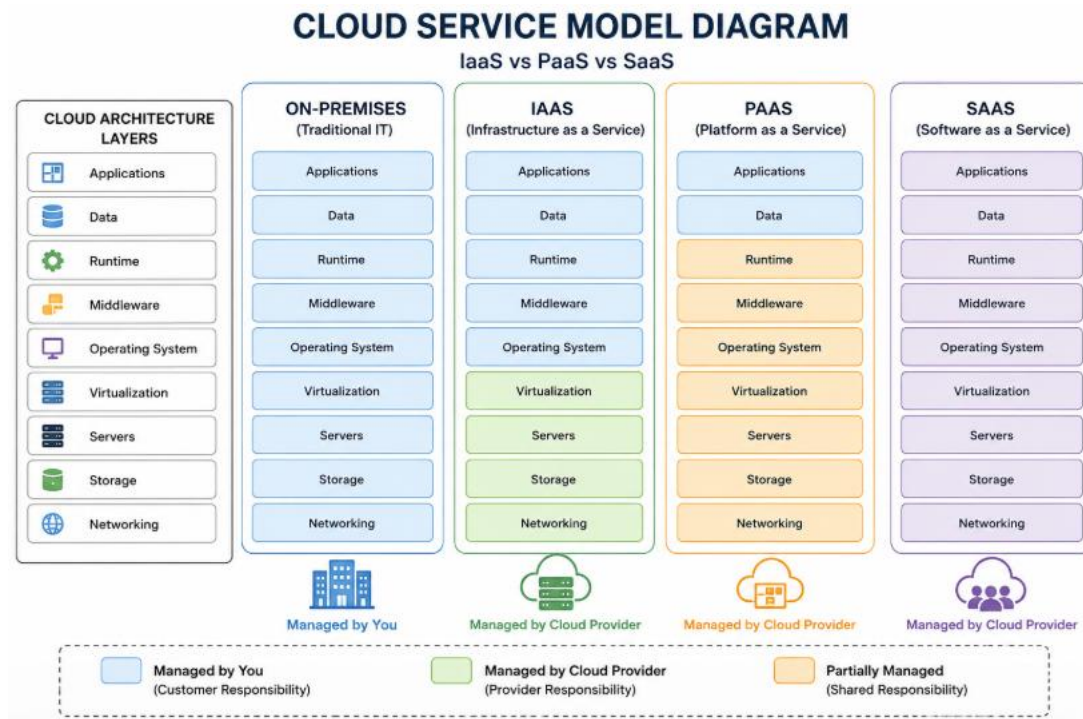
SYSTEM STUDY

Literature Review:-

Cloud security has become a major research area due to increasing cyber threats in cloud environments. Several researchers have identified misconfiguration, weak access control, and insecure APIs as primary causes of cloud security incidents.

Studies by IBM and Gartner highlight the importance of identity management, encryption, and continuous monitoring in cloud security.

AWS provides multiple cloud security services that help organizations secure their infrastructure through logging, monitoring, access management, and threat detection.



Comparative Analysis:-

Feature	AWS	Azure	Google Cloud
IAM	Advanced	Advanced	Moderate
Threat Detection	GuardDuty	Defender	Security Command Center
Monitoring	CloudTrail	Azure Monitor	Cloud Logging
Encryption	Supported	Supported	Supported

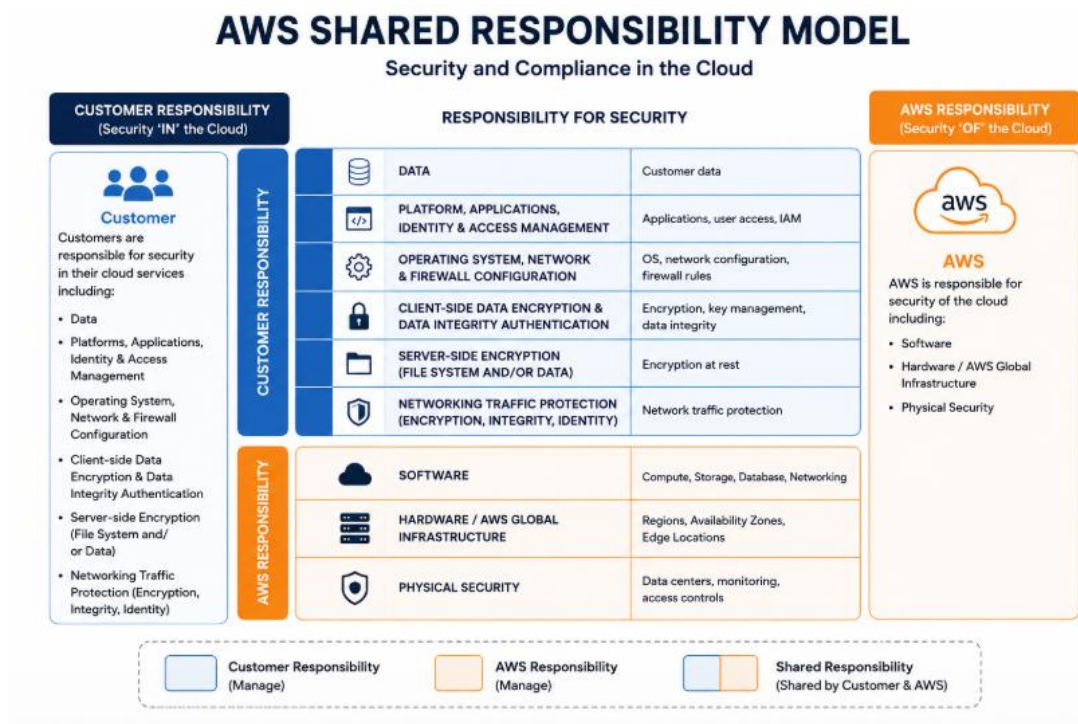
Security Frameworks:-

Shared Responsibility Model: - AWS secures cloud infrastructure while customers secure their applications, data, and access management.

Zero Trust Architecture: - Every user and device must be continuously verified before granting access.

IAM Framework: - Role-based access control ensures least privilege access.

Encryption Framework: - Encryption protects data both at rest and during transmission.



Research Gap:-

Most existing research focuses on theoretical cloud security concepts rather than practical implementation using real cloud platforms.

This project bridges the gap by implementing practical cloud security monitoring and risk mitigation using AWS

SYSTEM ANALYSIS

Functional Requirements:-

Requirement ID	Requirement
FR001	User authentication
FR002	Log monitoring
FR003	Threat detection
FR004	Access management
FR005	Security auditing

Non-Functional Requirements:-

Requirement	Description
Security	Secure cloud access
Scalability	Cloud scalability
Reliability	Continuous monitoring
Availability	High availability

User Requirements:-

Users require:

- Secure cloud access
- Threat visibility
- Activity monitoring
- Compliance tracking

Feasibility Study:-

Technical Feasibility: - AWS provides all required cloud security services.

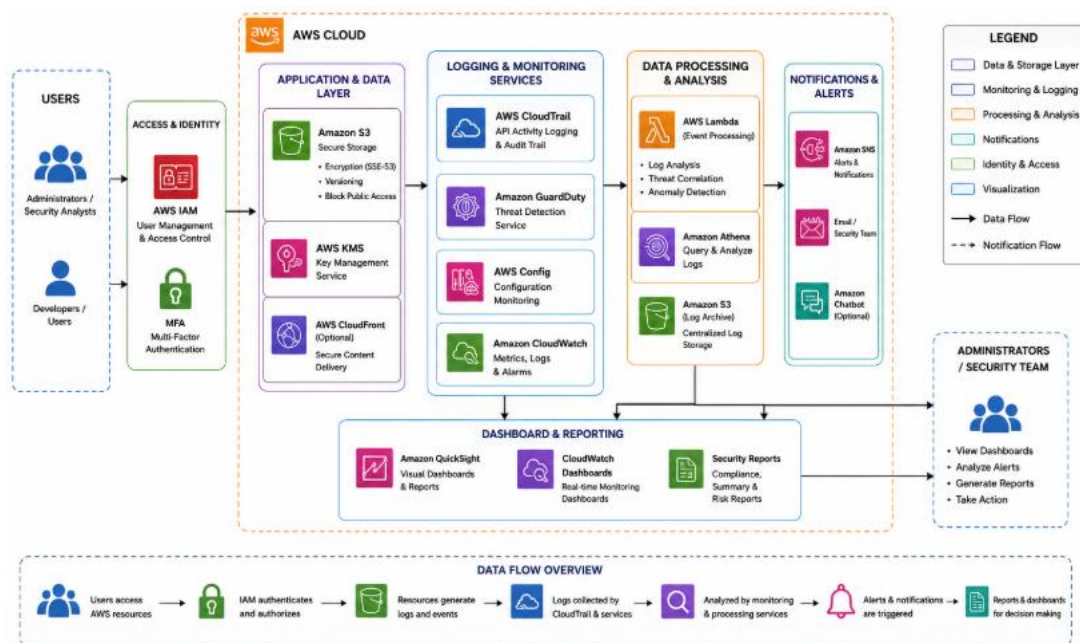
Economic Feasibility: - The project uses AWS free-tier services.

Operational Feasibility: - The system is easy to monitor and manage.

System Architecture:-

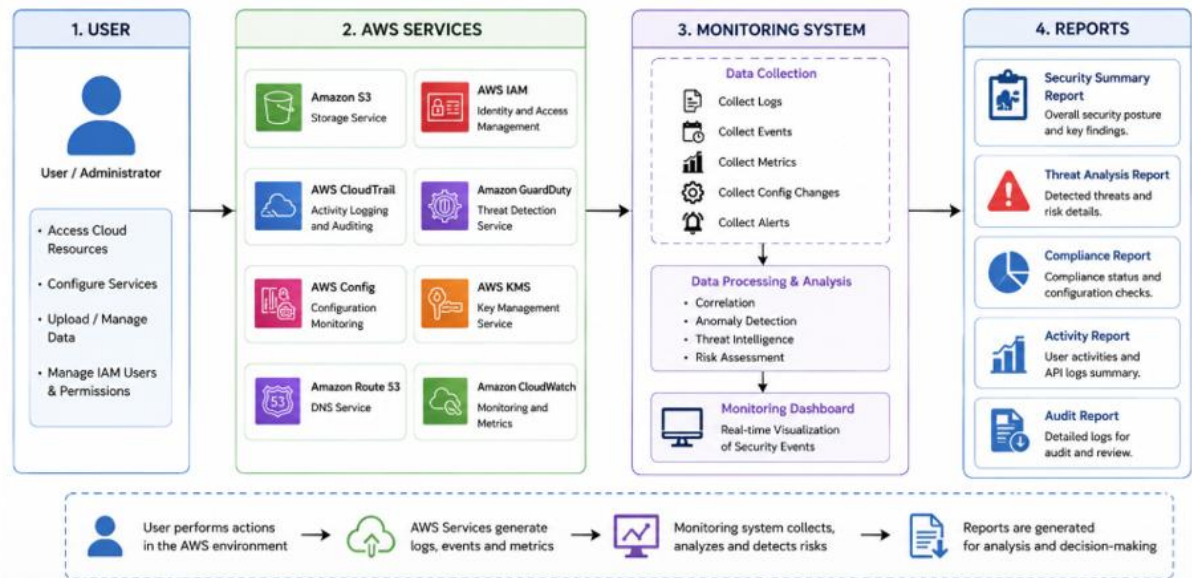
The system architecture consists of:

- Users
- AWS IAM
- S3 Storage
- CloudTrail Logging
- GuardDuty Monitoring
- AWS Config Compliance Monitoring



DFD (Data Flow Diagram) Level 0:-

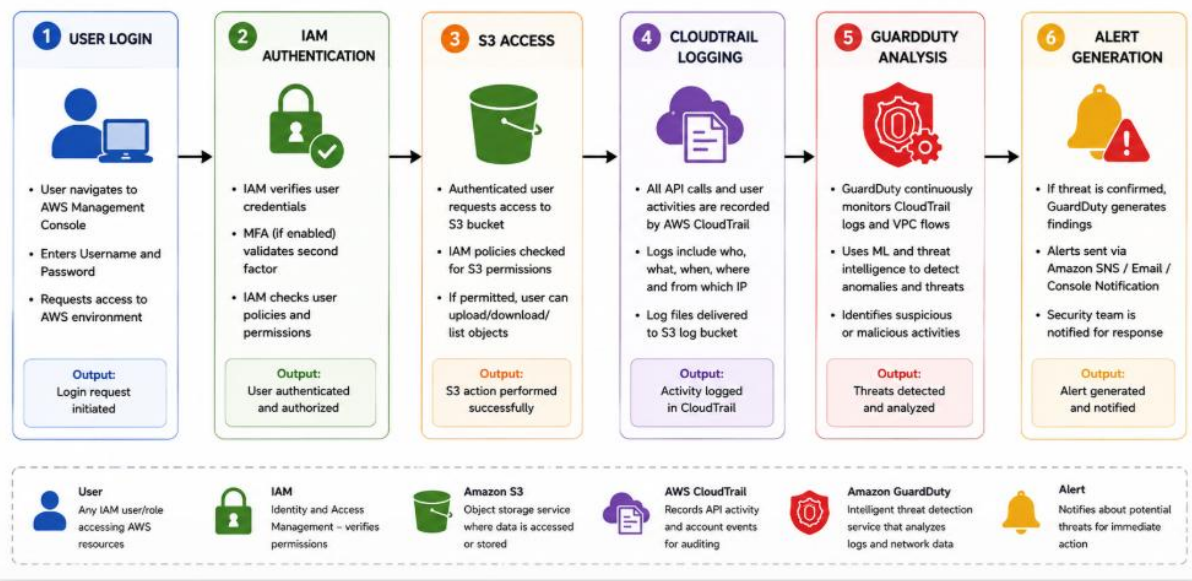
Cloud Security Analysis Flow



DFD Level 1:-

Cloud Security Monitoring Flow

Detailed Flow: User Login → IAM Authentication → S3 Access → CloudTrail Logging → GuardDuty Analysis → Alert Generation



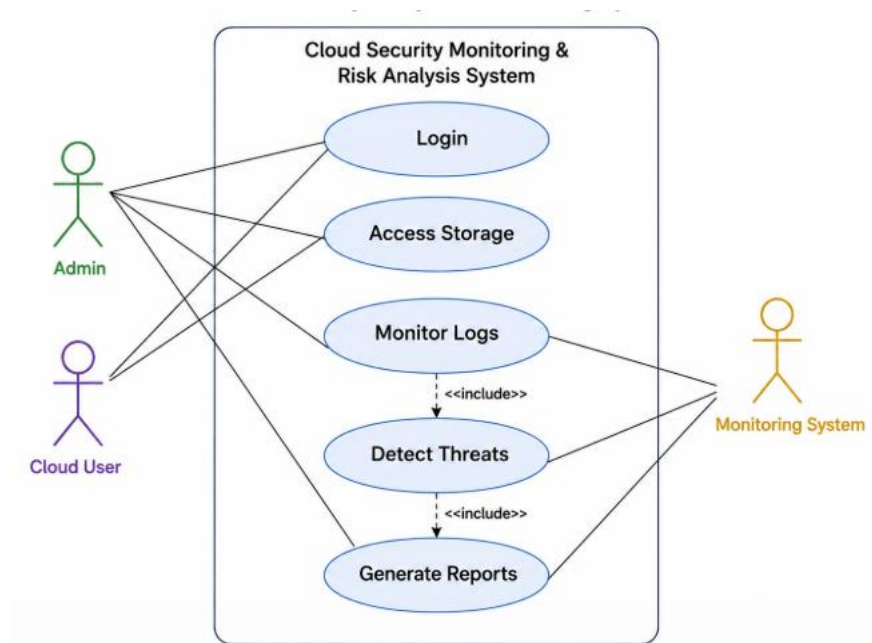
Use Case:-

Actors:

- Admin
- Cloud User
- Monitoring System

Use Cases:

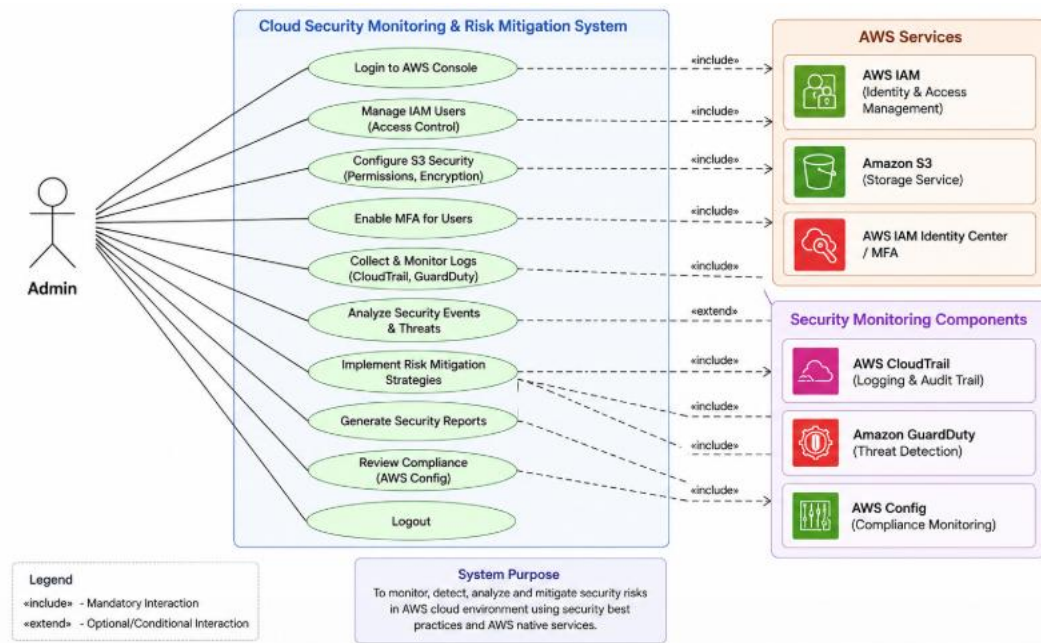
- Login
- Access Storage
- Monitor Logs
- Detect Threats
- Generate Reports



SYSTEM DESIGN

Use Case:-

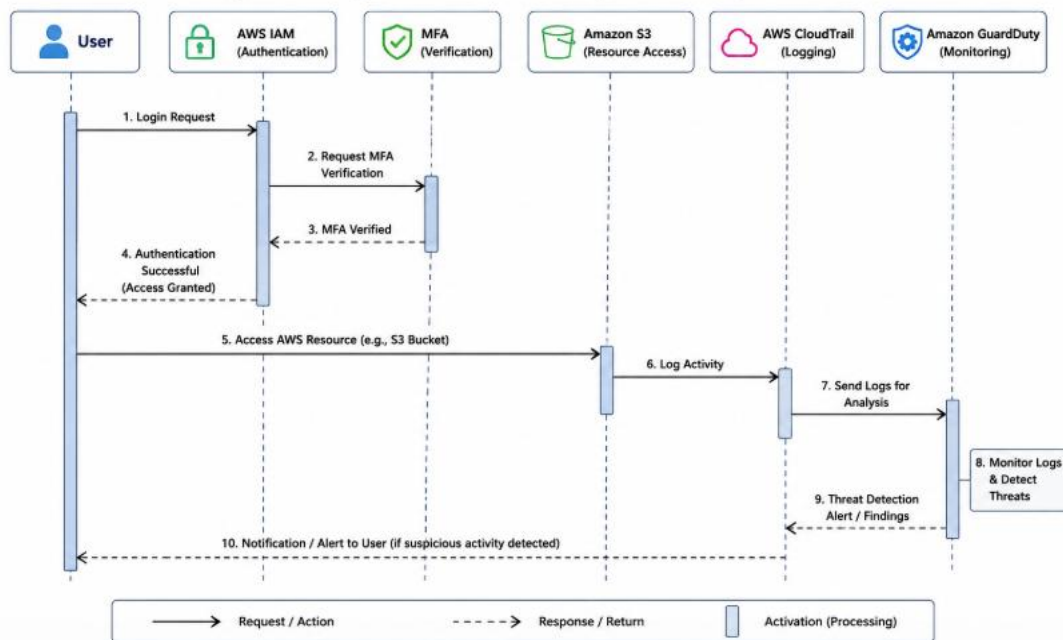
Cloud Security Monitoring & Risk Mitigation System (AWS)



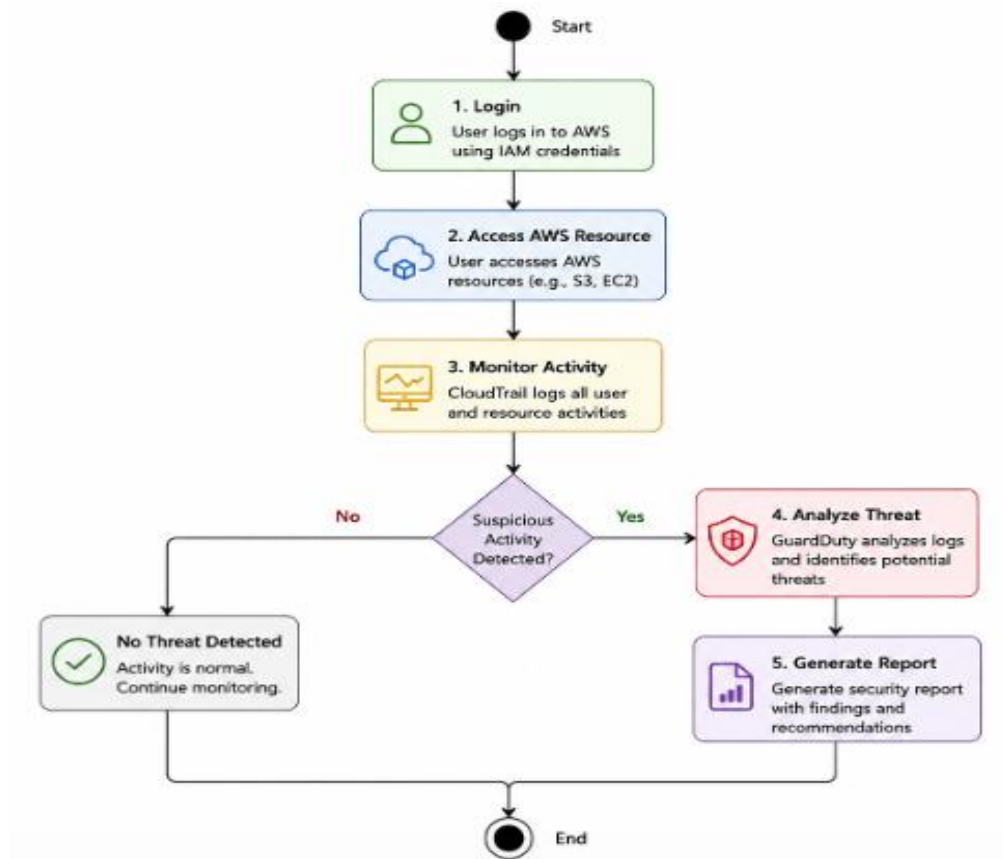
Classes:-

- **User Class:** Manages user authentication and access to AWS cloud resources.
- **IAM Class:** Handles identity management, user permissions, roles, and access control policies.
- **S3 Bucket Class:** Manages cloud storage operations, bucket security, encryption, and data access permissions.
- **Monitoring Class:** Monitors cloud activities, logs, and security events using CloudTrail and AWS Config.
- **Threat Detection Class:** Detects suspicious activities and security threats using GuardDuty and monitoring services.

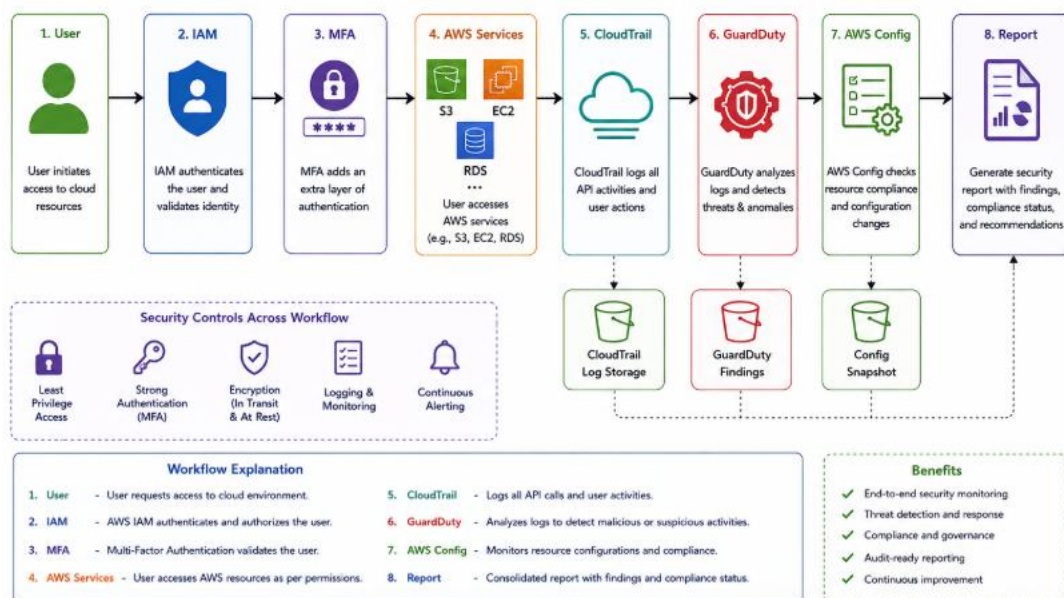
Sequence- Cloud Security Monitoring Flow:-



Activity Diagram:- Cloud Security Monitoring Flow



Security Workflow Design for Cloud Environment:-



SYSTEM IMPLEMENTATION

Development Environment:-

Component	Details
Platform	AWS Cloud
OS	Windows 11
Browser	Google Chrome
Cloud Services	AWS

AWS Services Used:-

Service	Purpose
Amazon S3	Secure cloud storage
AWS IAM	Access management
AWS CloudTrail	Activity logging
GuardDuty	Threat detection
AWS Config	Compliance monitoring

Hardware and Software Requirements:-

Hardware Requirements

- Intel Processor
- 8 GB RAM
- Internet connection

Software Requirements

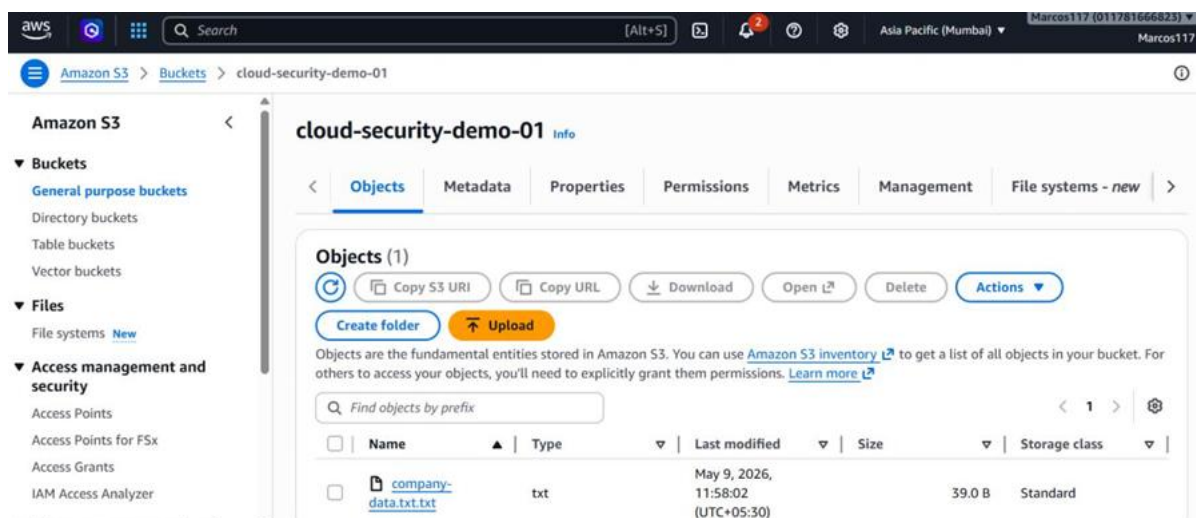
- AWS Account
- Browser
- MFA Application

Amazon S3 Implementation:-

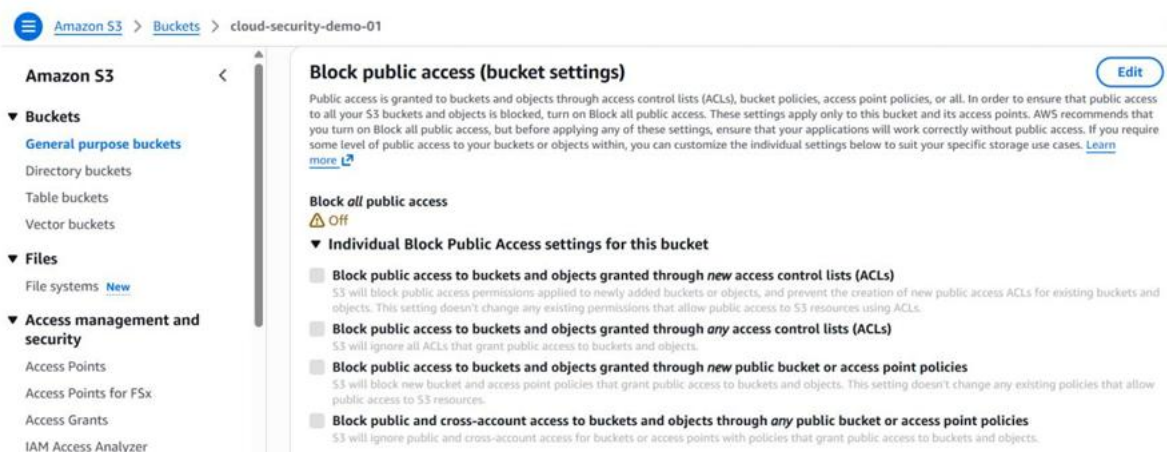
Amazon S3 bucket was created to demonstrate secure and insecure cloud storage configurations.

Implementation steps:

- Created S3 bucket
- Enabled public access (testing)
- Blocked public access
- Enabled encryption
- Enabled versioning



Bucket Permission:-



Amazon S3 > Buckets > cloud-security-demo-01

Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#) [View analyzer for ap-south-1](#)

Block public access (bucket settings) Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
On

► Individual Block Public Access settings for this bucket

Bucket Versioning Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Enabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)
Disabled

Default encryption Edit

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)
Server-side encryption with Amazon S3 managed keys (SSE-S3)

Bucket Key
When KMS encryption is used to encrypt new objects in this bucket, the bucket key reduces encryption costs by lowering calls to AWS KMS. [Learn more](#)
Enabled

Blocked encryption types [Info](#)
Server-Side Encryption with Customer-Provided Keys (SSE-C)

In-progress change to default encryption
As announced in November 2025, we have started deploying an update that will block server-side encryption with customer-provided keys (SSE-C) by default for all new buckets and specific existing buckets. If you need to use SSE-C encryption, make sure that SSE-C is not selected under Blocked encryption types. [Learn more](#)

IAM Implementation:-

IAM users and groups were created to implement role-based access control.

Implementation steps:

- Created IAM user
- Assigned AdministratorAccess
- Applied ReadOnlyAccess
- Created IAM group
- Enabled MFA

demo-user Info Delete

Summary

ARN arn:aws:iam::011781666823:user/demo-user	Console access Enabled with MFA	Access key 1 Create access key
Created May 09, 2026, 12:42 (UTC+05:30)	Last console sign-in Never	

demo-user

1 policy added

Permissions Groups (1) Tags Security credentials Last Accessed

Permissions policies (3) Remove Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

Search

Policy name	Type	Attached via
AdministratorAccess	AWS managed - job function	Directly
IAMUserChangePassword	AWS managed	Directly
ReadOnlyAccess	AWS managed - job function	Directly, Group SecurityAuditGroup

SecurityAuditGroup Info Delete

Summary Edit

User group name SecurityAuditGroup	Creation time May 09, 2026, 13:54 (UTC+05:30)	ARN arn:aws:iam::011781666823:group/SecurityAuditGroup
--	---	--

Users (1) Remove Add users

Users in this group (1)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

User name	Groups	Last activity	Creation time
demo-user	1	None	5 days ago

CloudTrail Implementation

CloudTrail was enabled to monitor AWS account activities.

Implementation steps:

- Created trail
- Enabled logging
- Generated events
- Monitored logs

CloudTrail

Trails

arn:aws:cloudtrail:ap-south-1:011781666823:trail/security-monitoring-trail

CloudTrail

Dashboard

Event coverage

Event history

Insights

Trails

Lake

Dashboards

Query

Event data stores

Integrations

Settings

security-monitoring-trail

Delete

Stop logging

Edit

General details

Trail logging

Logging

Trail log location

cloud-security-demo-01/AWSLogs/011781666823

Log file validation

Enabled

SNS notification delivery

Disabled

Trail name

security-monitoring-trail

Last log file delivered

May 09, 2026, 20:20:31 (UTC+05:30)

Last file validation delivered

May 09, 2026, 19:33:26 (UTC+05:30)

Last SNS notification

-

Multi-region trail

Yes

Log file SSE-KMS encryption

Enabled

Apply trail to my organization

Not enabled

AWS KMS key

arn:aws:kms:ap-south-1:011781666823:key/41f0917c-534f-4340-918d-0f1d9b2da11c

Event history (40)

Info

Download events

Query in Lake

Create Athena table

Event history shows you the last 90 days of management events.

Lookup attributes

Read-only

Q false

Filter by date and time

Clear filter

1

Event name	Event time	User name	Event source	Resource type
AttachUserPolicy	May 14, 2026, 21:26:39 (UTC+0...)	root	iam.amazonaws.com	AWS::IAM::User, AV
ConsoleLogin	May 14, 2026, 21:18:42 (UTC+0...)	root	signin.amazonaws.com	-
ConsoleLogin	May 14, 2026, 21:18:15 (UTC+0...)	root	signin.amazonaws.com	-

ConsoleLogin

Info

Details

Info

Event time

May 14, 2026, 21:18:42 (UTC+05:30)

AWS access key

-

AWS region

us-east-1

User name

root

Source IP address

157.33.205.4

Error code

-

Event name

ConsoleLogin

Event ID

4884f825-d05e-4a12-8fd7-9a705d46e1a3

Read-only

false

Event source

signin.amazonaws.com

Request ID

-

CloudTrail

Event history

AttachUserPolicy

CloudTrail

Dashboard

Event coverage

Event history

Insights

Trails

Lake

Dashboards

Query

AttachUserPolicy

Info

Details

Info

Event time

May 14, 2026, 21:26:39 (UTC+05:30)

AWS access key

ASIAQFPR57ADROSXCS22

AWS region

us-east-1

User name

root

Source IP address

157.33.205.4

Error code

-

Event name

AttachUserPolicy

Event ID

af63dbcc-5ee3-466a-bfd1-89cc00cc30a2

Read-only

false

Event source

iam.amazonaws.com

Request ID

688fd499-94b3-48dd-add3-65cc85ff96b7

GuardDuty Implementation:-

GuardDuty was enabled for threat monitoring and suspicious activity detection.

GuardDuty > Summary

Summary Info

Updated a few seconds ago Today ▾

View and analyze security trends based on GuardDuty findings in your AWS environment.

Overview

Attack sequences - <i>new</i>	Total findings	Resources with findings	Accounts with findings
0	1	1	1

GuardDuty > Findings

Findings (1/1) Info

Create suppression rule Actions ▾

Saved filters Apply saved filters ▾ Status Current ▾ Threat type All findings ▾ < 1 >

☒	Title	Severity ▾	Finding type ▾
☒	The API ListManagedNotificationEvents was invoked using root credentials.	Low	Policy:IAMUser/RootCredentialUsage

AWS Config Implementation:-

AWS Config was enabled to monitor compliance and configuration changes.

AWS Config > Dashboard

Compliance status

Rules	Resources
0 Noncompliant rule(s)	0 Noncompliant resource(s)
0 Compliant rule(s)	0 Compliant resource(s)

TESTING

Testing Strategy:-

The project was tested using functional testing, security testing, and integration testing.

Unit Testing:-

Individual AWS services were tested separately.

Integration Testing:-

Integration between IAM, CloudTrail, GuardDuty, and S3 was verified.

System Testing:-

Complete workflow testing was performed.

Test Cases:-

Test Case ID	Input	Expected Output	Actual Output	Status
TC001	Public S3 Access	Access Denied	Access Denied	Pass
TC002	Login Without MFA	Authentication Failure	Authentication Failure	Pass
TC003	IAM Policy Change	Event Logged	Logged Successfully	Pass
TC004	CloudTrail Monitoring	Activity Recorded	Recorded	Pass

RESULTS AND DISCUSSION

Security Threat Analysis:-

The project successfully identified major cloud security threats including:

- Public cloud storage exposure
- Excessive IAM permissions
- Weak authentication
- Lack of monitoring

Risk Mitigation Results:-

Implementation of security controls improved cloud security posture through:

- Encryption
- MFA
- Access restriction
- Continuous monitoring

Performance Evaluation:-

The system improved:

- Security visibility
- Threat monitoring
- Access management
- Compliance tracking

Risk Analysis Table:-

Activity	Risk	Severity
Public S3 Access	Data Exposure	High
Excessive IAM Access	Unauthorized Access	High

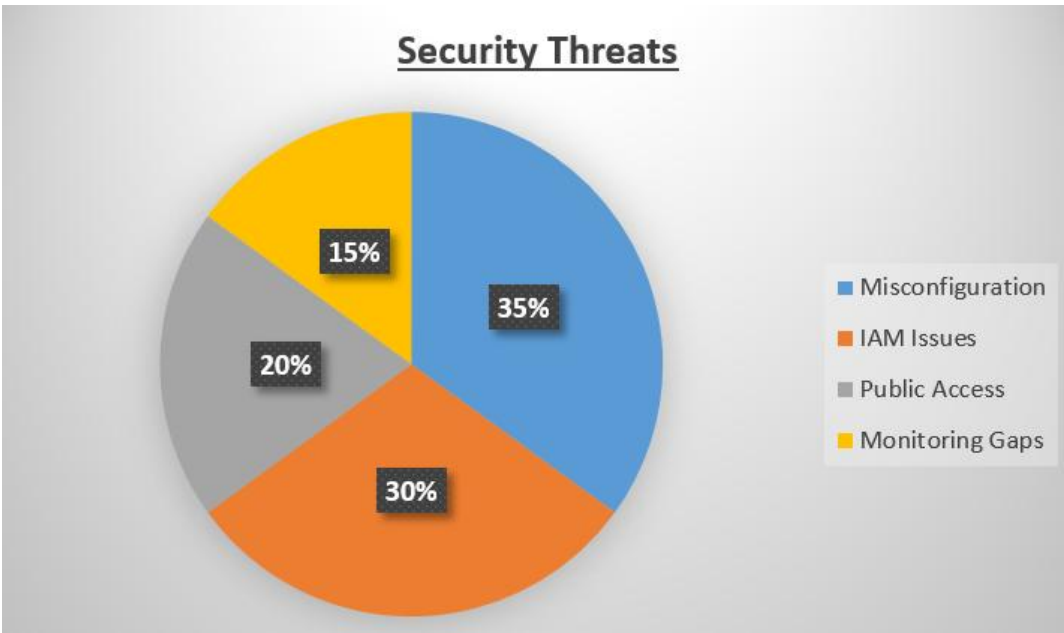
Activity	Risk	Severity
Frequent Login Attempts	Credential Misuse	Medium
Permission Changes	Misconfiguration	Medium

Graphical Analysis

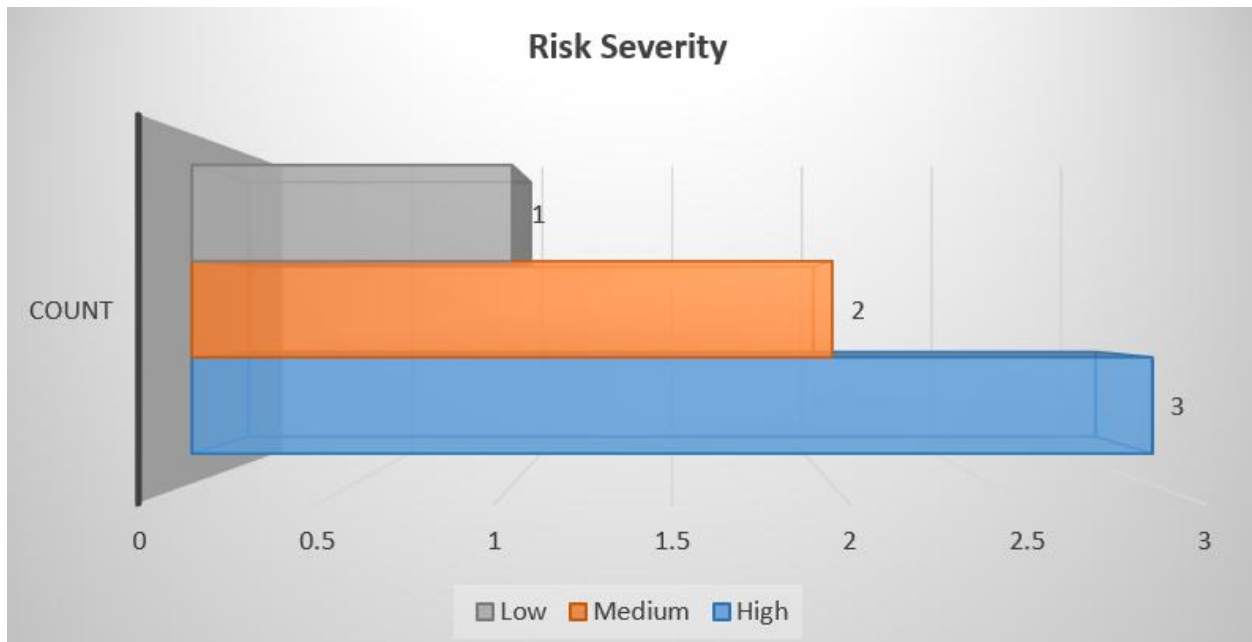
Pie Chart :- Threat Distribution Analysis

Bar Graph :- Risk Severity Analysis

Distribution of Security Threats by my own analysis:-



Risk Severity by my own analysis:-



Cloudtrail Analysis - Before:

Active Trail:

CloudTrail > Trails > arn:aws:cloudtrail:ap-south-1:011781666823:trail/security-monitoring-trail

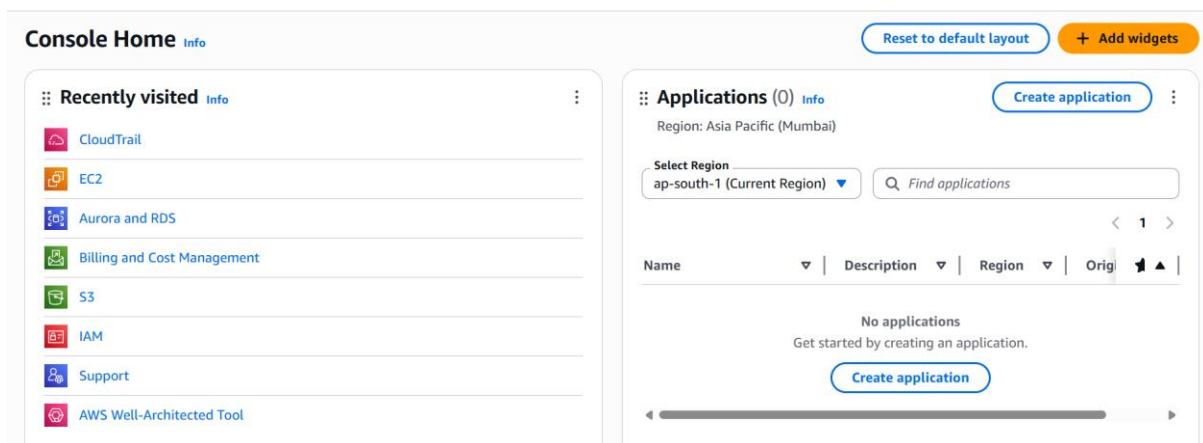
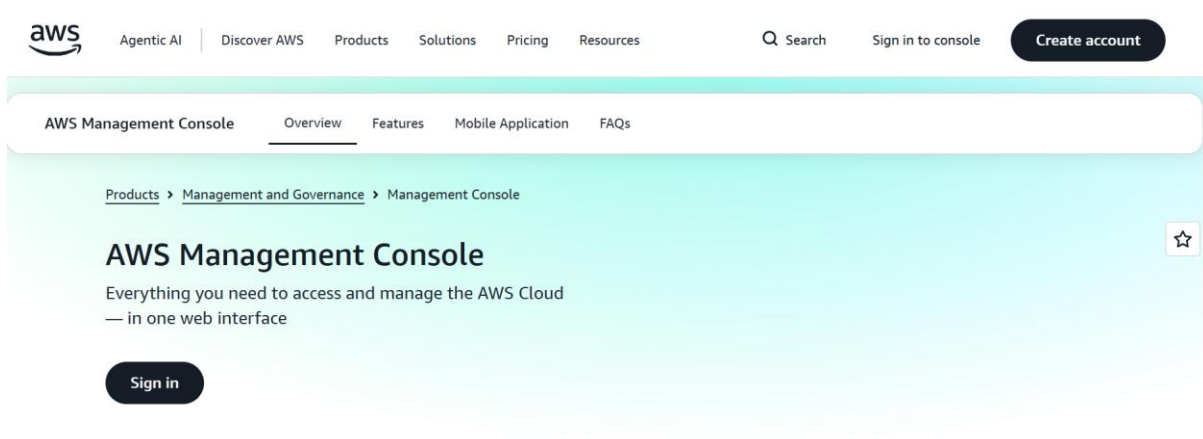
security-monitoring-trail

[Delete](#) [Stop logging](#) [Edit](#)

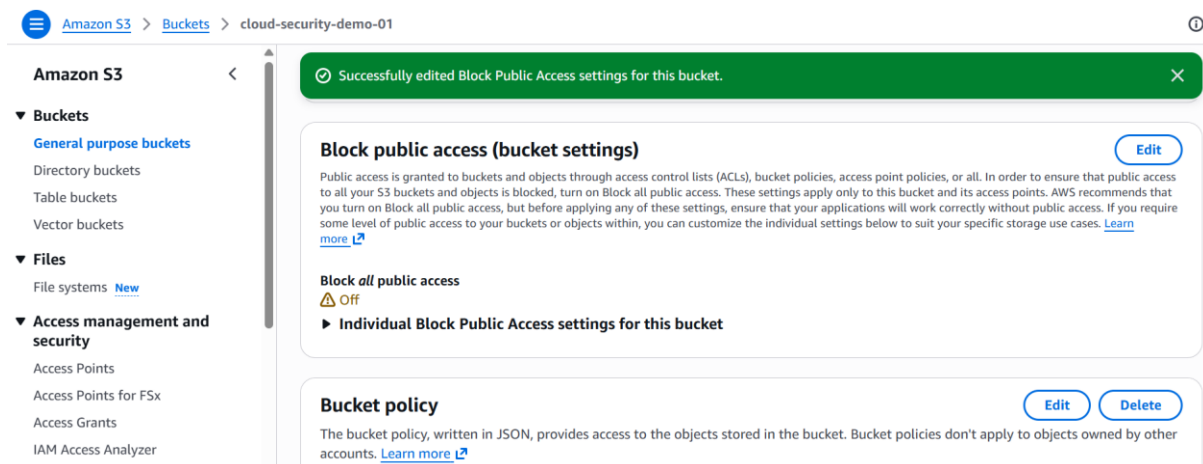
General details	Trail log location cloud-security-demo-01/AWSLogs/011781666823	Log file validation Enabled	SNS notification delivery Disabled
Trail logging Logging	Last log file delivered May 09, 2026, 20:20:31 (UTC+05:30)	Last file validation delivered May 09, 2026, 19:33:26 (UTC+05:30)	Last SNS notification -
Trail name security-monitoring-trail	Log file SSE-KMS encryption Enabled		
Multi-region trail Yes	AWS KMS key arn:aws:kms:ap-south-1:011781666823:key/41f0917c-534f-4340-918d-0f1d9b2da11c		
Apply trail to my organization Not enabled			

Performed Activities:

Login and Logout from account



Edit S3 Bucket Permission



Amazon S3 > Buckets > cloud-security-demo-01

Amazon S3

- Buckets**
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- Files**
 - File systems [New](#)
- Access management and security**
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer

Successfully edited Block Public Access settings for this bucket.

Block public access (bucket settings) [Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
☒ On

► **Individual Block Public Access settings for this bucket**

Bucket policy [Edit](#) [Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Create/delete IAM user

IAM > IAM users

Identity and Access Management (IAM)

Search IAM

Dashboard

- Access Management**
 - Roles
 - Policies
 - IAM users**
 - IAM user groups
 - Identity providers
 - Account settings
 - Root access management

User created successfully
 You can view and download the user's password and email instructions for signing in to the AWS Management Console.
[View user](#)

IAM users (1/3) [Info](#) [Refresh](#) [Delete](#) [Create user](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

	User name	Path	Group	Last activity	MFA	Password age
☐	demo-user	/	1	-	Virtual...	8 hours
☒	RAJ	/	0	-	-	-

IAM > IAM users

Identity and Access Management (IAM)

Search IAM

Dashboard

- Access Management**
 - Roles
 - Policies
 - IAM users**

User "RAJ" deleted.

IAM users (2) [Info](#) [Refresh](#) [Delete](#) [Create user](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

	User name	Path	Group	Last activity	MFA	Password age
☐	demo-user	/	1	-	Virtual...	8 hours

Stop DB -

Aurora and RDS > Databases

Dashboard
Databases
Query editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies

Your database **calc-db** is stopped temporarily. It will restart automatically after 7 days, on May 16, 2026, 20:37 (UTC+05:30)

Databases (1) Group resources Modify Actions Create database

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order
calc-db	Stopped...	Instance	MySQL Co...	SECOND

Stop EC2 Instance –

EC2 > Instances

Dashboard
AWS Global View
Events
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager

Instances (1/1) Last updated less than a minute ago Connect Instance state Actions Launch instances

Saved filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
CalcApp-EC2	i-0100fc6c3b023326b	Stopped	t2.micro	-	View alarms

i-0100fc6c3b023326b (CalcApp-EC2)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary

Instance ID i-0100fc6c3b023326b	Public IPv4 address -	Private IPv4 addresses 10.0.1.157
---	---------------------------------	---

Cloudtrail Analysis - After:-

CloudTrail > Event history > ConsoleLogin

Dashboard
Event coverage
Event history
Insights
Trails

ConsoleLogin

Details

Event time May 09, 2026, 21:03:11 (UTC+05:30)	AWS access key -	AWS region us-east-1
User name root	Source IP address 157.32.138.226	Error code -
Event name ConsoleLogin	Event ID 36727c4e-83e9-4c07-b889-f40657cbd63a	Read-only false
Event source signin.amazonaws.com	Request ID -	

Enable MFA

CloudTrail

Event history

EnableMFADevice

CloudTrail

Dashboard

Event coverage

Event history

Insights

Trails

Lake

Dashboards

Query

EnableMFADevice

Details

Event time

May 09, 2026, 14:01:12 (UTC+05:30)

User name

root

Event name

EnableMFADevice

Event source

iam.amazonaws.com

AWS access key

ASIAQFPR57AD2S4ENXXW

Source IP address

157.32.117.201

Event ID

246e6803-1044-4e9a-85c5-f3ef7d4a1e7b

Request ID

e70cf73e-dc56-4d3d-bea1-4544658b0ce7

AWS region

us-east-1

Error code

-

Read-only

false

Stop Instance

CloudTrail

Event history

StopInstances

CloudTrail

Dashboard

Event coverage

Event history

Insights

Trails

Lake

Dashboards

Query

StopInstances

Details

Event time

May 09, 2026, 20:32:24 (UTC+05:30)

User name

root

Event name

StopInstances

Event source

ec2.amazonaws.com

AWS access key

ASIAQFPR57ADSFKEFBM7

Source IP address

157.32.138.226

Event ID

46472453-f1db-4948-bc76-87b8f424b46c

Request ID

10e0552a-b22c-4427-a0c0-1d902efb7636

AWS region

ap-south-1

Error code

-

Read-only

false

S3-Block Public Access

CloudTrail

Event history

PutBucketPublicAccessBlock

CloudTrail

Dashboard

Event coverage

Event history

Insights

Trails

Lake

Dashboards

Query

PutBucketPublicAccessBlock

Details

Event time

May 09, 2026, 21:09:04 (UTC+05:30)

User name

root

Event name

PutBucketPublicAccessBlock

Event source

s3.amazonaws.com

AWS access key

ASIAQFPR57AD6CQLKARK

Source IP address

157.32.138.226

Event ID

2adff22c-4b5d-4107-a6a4-a92e9f983e5b

Request ID

668N9X2530KFYW6D

AWS region

ap-south-1

Error code

-

Read-only

false

Create/Delete User

CloudTrail

Event history

CreateUser

CloudTrail

Dashboard

Event coverage

Event history

Insights

Trails

Lake

Dashboards

Query

CreateUser

Details

Event time

May 09, 2026, 21:16:45 (UTC+05:30)

User name

root

Event name

CreateUser

Event source

iam.amazonaws.com

AWS access key

ASIAQFPR57ADQQ3TVV7B

Source IP address

157.32.138.226

Event ID

d188d4d9-d6fc-467a-8a47-176c1fded052

Request ID

cc31ed39-40dd-46f2-87b6-e30996c03cb8

AWS region

us-east-1

Error code

-

Read-only

false

CloudTrail

Dashboard
Event coverage
Event history
Insights
Trails

▼ Lake

Dashboards
Query
Event data stores
Integrations

Settings

Details [Info](#)

Event time May 09, 2026, 21:17:41 (UTC+05:30)	AWS access key ASIAQFPR57ADQQ3TVV7B	AWS region us-east-1
User name root	Source IP address 157.32.138.226	Error code -
Event name DeleteUser	Event ID 494bb20a-4659-4db8-a758-cd5609eb9b8c	Read-only false
Event source iam.amazonaws.com	Request ID bb127931-5e68-4d1b-b89f-66530d099e9e	

Resources referenced (1) [Info](#)

Resources referenced describes the name or ID of resources that were read or changed by an event

Resource type	Resource name	AWS Config resource timeline
AWS::IAM::User	RAJ	Enable AWS Config resource recording

CONCLUSION AND FUTURE SCOPE

Conclusion:-

The project successfully analysed cloud security challenges and implemented practical risk mitigation strategies using AWS cloud services.

The implementation demonstrated:

- Secure cloud storage
- Identity and access management
- Threat monitoring
- Logging and compliance management
- Security testing and analysis

The project achieved all defined objectives and improved understanding of practical cloud security implementation.

Future Scope:-

Future improvements may include:

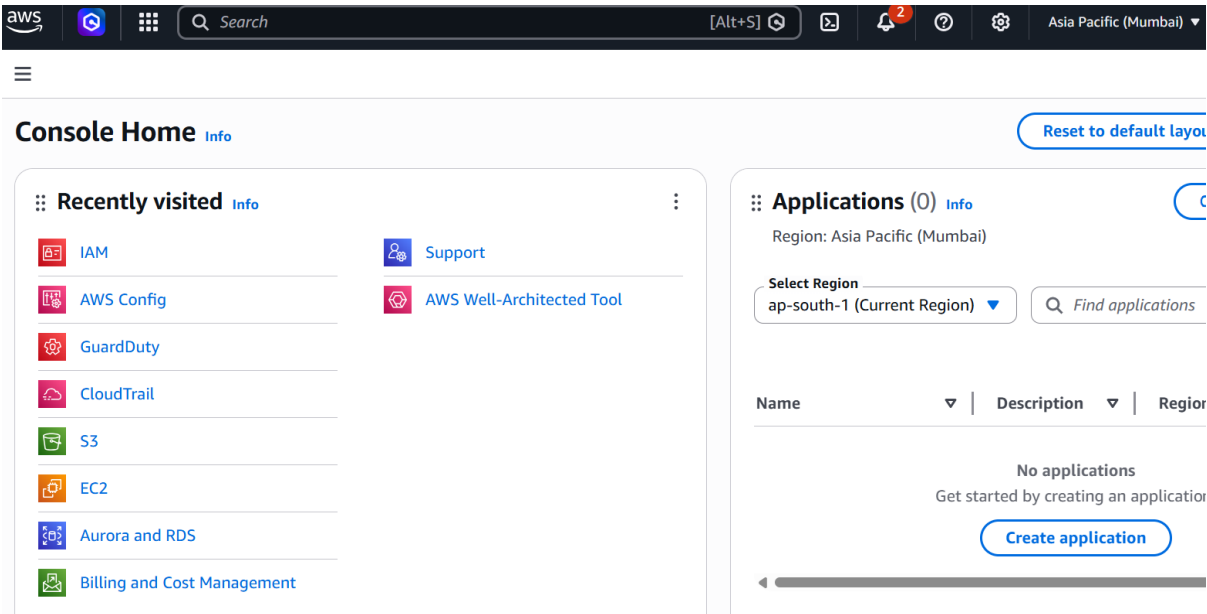
- AI-based threat detection
- SIEM integration
- Multi-cloud monitoring
- Automated remediation
- Advanced Zero Trust implementation
- Mobile monitoring dashboard

REFERENCES

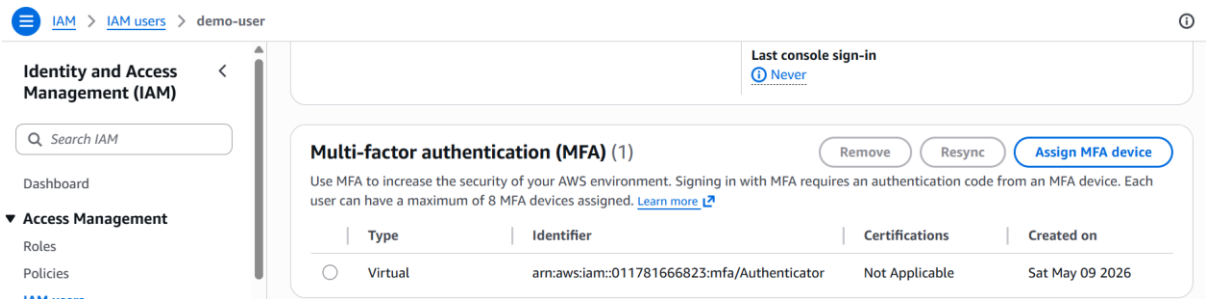
1. Amazon Web Services Documentation
2. IBM Cloud Security Report
3. Gartner Cloud Security Research
4. NIST Cloud Computing Guidelines
5. IEEE Research Papers on Cloud Security
6. AWS IAM Documentation
7. AWS CloudTrail Documentation
8. AWS GuardDuty Documentation
9. AWS Config Documentation
10. Cloud Security Alliance Reports
11. Journal of Cloud Computing
12. Google Cloud Security Whitepapers
13. Microsoft Azure Security Documentation
14. Research Papers on Zero Trust Architecture
15. Research Papers on Cloud Encryption
16. Cybersecurity Journals
17. International Journal of Information Security
18. AWS Well-Architected Framework
19. DevSecOps Research Papers
20. AI-based Threat Detection Research Articles

APPENDICES

Appendix A: Additional AWS Screenshots – Console



IAM - MFA



S3 Policy

Amazon S3

▼ Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

▼ Files

File systems [New](#)

▼ Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems [New](#)

Access management and security

Access Points

Access Points for FSx

Access Grants

Public access is blocked because Block Public Access settings are turned on for this bucket

To determine which settings are turned on, check your Block Public Access settings for this bucket. [Learn more about Amazon S3 Block Public Access](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AWSCloudTrailAclCheck20150319-60fe97cc-4d27-4193-8e66-4af07fbf7a32",
      "Effect": "Allow",
      "Principal": {
        "Service": "cloudtrail.amazonaws.com"
      },
      "Action": "s3:GetBucketAcl",
      "Resource": "arn:aws:s3:::cloud-security-demo-01",
      "Condition": {
        "StringEquals": {
          "AWS:SourceArn": "arn:aws:cloudtrail:ap-south-1:011781666823:trail/security-monitoring-trail"
        }
      }
    },
    {
      "Sid": "AWSCloudTrailWrite20150319-81d56097-67b9-4820-a598-3653e2dc0961",
      "Effect": "Allow",
      "Principal": {
        "Service": "cloudtrail.amazonaws.com"
      },
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3:::cloud-security-demo-01/AWSLogs/011781666823/*",
      "Condition": {
        "StringEquals": {
          "AWS:SourceArn": "arn:aws:cloudtrail:ap-south-1:011781666823:trail/security-monitoring-trail",
          "s3:x-amz-acl": "bucket-owner-full-control"
        }
      }
    }
  ]
}
```

Amazon S3

▼ Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

▼ Files

File systems [New](#)

▼ Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Access control list (ACL)

Edit

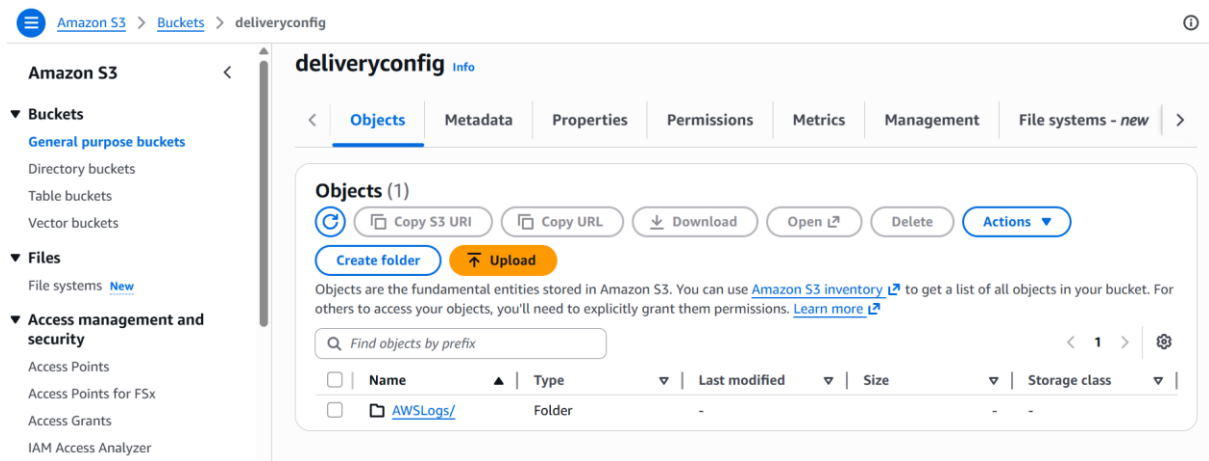
Grant basic read/write permissions to other AWS accounts. [Learn more](#)

This bucket has the bucket owner enforced setting applied for Object Ownership

When [bucket owner enforced](#) is applied, use bucket policies to control access. [Learn more](#)

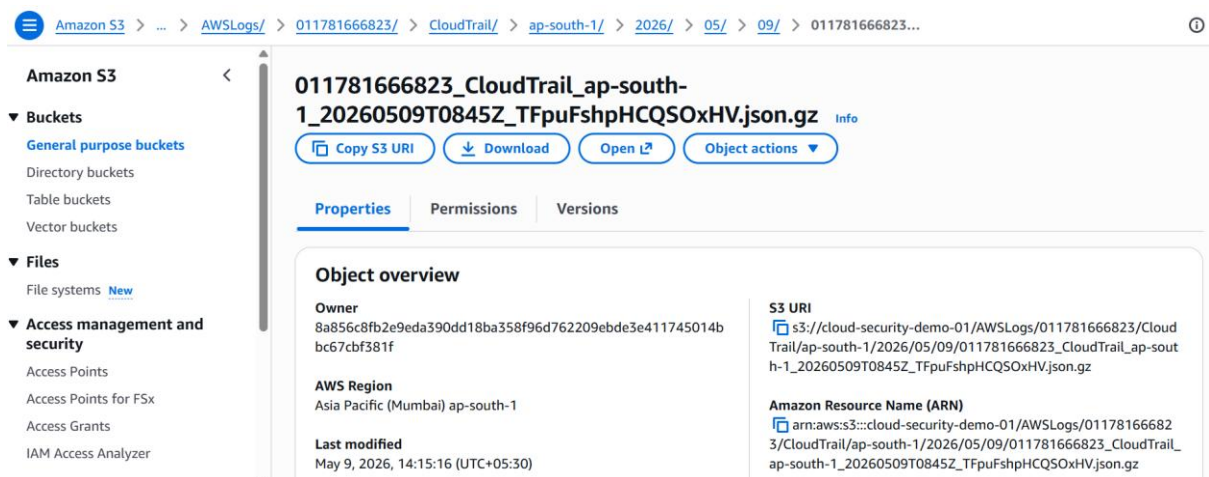
Grantee	Objects	Bucket ACL
Bucket owner (your AWS account) Canonical ID: 8a856c8fb2e9eda390dd18ba358f96d762209ebde3e411745014bbc67cbf381f	List, Write	Read, Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	-	-
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	-	-
S3 log delivery group Group: http://acs.amazonaws.com/groups/s3/LogDelivery	-	-

Config-logs S3 Bucket



Appendix B: Sample Logs

Cloudfaril Raw logs



Event History Table

Event name	Event time	User name	Event source	Resource type	Resource name
DescribeDomainConfiguration	May 14, 2026, 22:00:11 (UTC+05:30)	ConfigResourceCompositionSession	iot.amazonaws.com	-	-
DescribeDomainConfiguration	May 14, 2026, 22:00:11 (UTC+05:30)	ConfigResourceCompositionSession	iot.amazonaws.com	-	-

ListDomainConfigurations	May 14, 2026, 22:00:11 (UTC+05:30)	ConfigResourceCompositionSession	iot.amazonaws.com	-	-
StartConfigurationRecorder	May 14, 2026, 21:39:57 (UTC+05:30)	root	config.amazonaws.com	-	-
PutDeliveryChannel	May 14, 2026, 21:39:56 (UTC+05:30)	root	config.amazonaws.com	-	-
PutConfigurationRecorder	May 14, 2026, 21:39:53 (UTC+05:30)	root	config.amazonaws.com	-	-
AttachUserPolicy	May 14, 2026, 21:26:39 (UTC+05:30)	root	iam.amazonaws.com	AWS::IAM::User, AWS::IAM::Policy	demo-user, arn:aws:iam::aws:policy/AdministratorAccess
ConsoleLogin	May 14, 2026, 21:18:42 (UTC+05:30)	root	signin.amazonaws.com	-	-

Appendix C: Testing

The AWS cloud security environment was tested using authentication, monitoring, and security validation techniques.

Appendix D: User Manual

Steps:

1. Login to AWS
2. Access S3 Bucket
3. Configure IAM
4. Enable CloudTrail
5. Monitor GuardDuty
6. Review AWS Config

Appendix E: Installation Guide

1. Create AWS Account
2. Enable AWS Services
3. Configure IAM
4. Setup S3
5. Enable Monitoring
6. Configure Security Controls